

Uniform Partition Sampler: BE Distribution Verification

Summary of `mc_uniform_fast.py`

(a) Theory

The uniform measure on partitions

The script samples integer partitions of L under the **uniform measure**: every partition of L receives equal weight $1/p(L)$. This is the canonical ensemble of Assumption 1 in the paper (Section 1), under which the run-length marginal distribution converges to the Bose-Einstein (BE) form rather than the geometric distribution that arises in the composition (ordered) ensemble.

Bose-Einstein run-length distribution

Under the uniform measure the expected frequency of parts of size j is

$$f_{\text{BE}}(j) = \frac{1}{E[k]_{\text{canon}}} \cdot \frac{1}{e^{j\alpha} - 1}, \quad \alpha = \frac{\pi}{\sqrt{6L}},$$

where $E[k]_{\text{canon}}$ is the canonical mean part count. The fugacity α is fixed not by $\pi/\sqrt{6L}$ directly but by solving the self-consistency equation $\sum_{j \geq 1} (e^{j\alpha} - 1)^{-1} = E[k]_{\text{canon}}$ for the exact DP canonical mean, which avoids the error that arises from using the grand canonical normaliser $E[k]_{\text{GC}} = (\pi/\sqrt{6})\sqrt{L}$ instead. The two normalisers differ by a factor of order $\log L$ (Section 6.3 of the paper), and using the GC normaliser produces a distribution that does not sum to 1.

The competing geometric distribution has the same mean run length $\bar{r} = L/E[k]_{\text{canon}}$ but replaces $1/(e^{j\alpha} - 1)$ with $e^{-j\alpha}/(1 - e^{-\alpha})$, which underestimates $f(j)$ at small j and overestimates the tail. The bosonic enhancement factor $(1 - e^{-j\alpha})^{-1}$ grows without bound as $L \rightarrow \infty$ for fixed j , so the two distributions become increasingly distinguishable at large L .

Canonical mean part count and k^* scaling

Under the uniform measure the modal and mean part counts scale as

$$k^* \sim 0.950 \cdot \frac{\sqrt{6}}{2\pi} \sqrt{L} \log L, \quad E[k]_{\text{canon}} \sim \frac{\sqrt{6}}{2\pi} \sqrt{L} \log L,$$

by the Erdős–Lehner theorem. The naive $\sqrt{L}$ estimate $(\pi/\sqrt{6})\sqrt{L}$ underestimates both by a factor of order $\log L$, growing from approximately $2\times$ at $L = 100$ to $3\times$ at $L = 2000$.

(b) What the script does

The script has two operating modes.

Standalone mode (`python3 mc_uniform_fast.py`) runs four tasks sequentially:

1. **Validation:** compares the MC distribution against the exact uniform distribution for $L \leq 20$, reporting total variation distance and error in $\langle k \rangle$.
2. **Benchmark:** measures steps/second at representative L values.
3. **Mode convergence study:** estimates the modal k^* and mean $\langle k \rangle$ under the uniform measure for $L \in \{20, 50, 100, 200, 500, 1000, 2000, 5000, 10000\}$, confirming the $\sqrt{L} \log L$ Erdős–Lehner scaling.
4. **Run-length distribution (RLD) study:** measures $f(j)$ under the uniform measure for $L \in \{100, 200, 500\}$, compares against the BE and geometric predictions, and computes KL divergences (Section 8.3 of the paper).

SLURM job-array mode (`--task i --outdir dir`) runs a single task (one L from the mode study or one L from the RLD study) and writes a JSON result file. This allows the full suite to run in parallel across SLURM array tasks; `mc_uniform.sh` configures the array and `aggregate.py` combines outputs.

(c) How the script does it

Partition representation

The script imports the numpy-accelerated partition infrastructure from `partition_numpy.py`. A partition of L into k parts is stored as two short `int32` arrays `vals` (distinct part values, ascending) and `mults` (corresponding multiplicities), with $\sum_i \text{vals}[i] \cdot \text{mults}[i] = L$. For a typical partition $n_d = O(\sqrt{L})$, so all operations run in $O(\sqrt{L})$ time.

MH step under the uniform measure

`metropolis_step_uniform_fast` selects a move type — transfer (probability $p_T = 0.5$), split (0.25), or merge (0.25) — then samples a move uniformly within that type. The acceptance ratio is

$$\alpha_{\text{MH}} = \min\left(1, \frac{N_{\text{fwd}}}{N_{\text{rev}}}\right),$$

where N_{fwd} and N_{rev} are the number of available moves of the chosen type before and after the proposal. Since the target measure assigns equal weight to all partitions, the weight ratio is identically 1 and drops out; only the proposal asymmetry correction $N_{\text{fwd}}/N_{\text{rev}}$ remains. This is computed from the exact move counts supplied by `partition_numpy.py` without enumerating all moves.

The three move types together make the chain irreducible on the set of all partitions of L : transfer moves explore within fixed k , while split and merge moves change k by ± 1 and connect partitions of different part counts.

Equilibration and sampling

For the mode study, each L uses a burn-in of $\max(50,000, 300L)$ steps followed by $\max(1,000,000, 5,000L)$ thinned samples with thinning factor $\max(1, \lfloor L/20 \rfloor)$. For the RLD study, each L uses an equilibration of $n_{\text{steps}}/5$ steps (where $n_{\text{steps}} = 600,000$ by default) followed by n_{steps} sampling steps, accumulating part-value frequencies via a `Counter`.

BE and geometric reference distributions

The BE reference is parameterised by the canonical α , obtained by bisection on the self-consistency equation $\sum_{j \geq 1} (e^{j\alpha} - 1)^{-1} = E[k]_{\text{canon}}$ to bracket width below 10^{-10} , with the series truncated when terms fall below 10^{-12} . The exact DP canonical mean is hard-coded for $L \in \{100, 200, 500\}$: $E[k]_{\text{canon}} = 21.75, 34.17, 61.37$ respectively. The geometric reference uses the same mean run length $\bar{r} = L/E[k]_{\text{canon}}$.

KL divergences $D_{\text{KL}}(f_{\text{MC}} \| f_{\text{BE}})$ and $D_{\text{KL}}(f_{\text{MC}} \| f_{\text{geom}})$ are computed over all j where $f_{\text{MC}}(j) > 0$.

Validation

`validate` compares the MC distribution against the exact uniform distribution for even $L = 4\text{--}20$ using 2×10^5 samples after 5×10^4 equilibration steps, computing total variation distance and relative error in $\langle k \rangle$.

(d) Output produced

Validation

For each even L from 4 to 20, the table reports the exact modal k under the uniform measure, the Hardy–Ramanujan prediction $(\pi/\sqrt{6})\sqrt{L}$, the exact mean $\langle k \rangle_{\text{exact}}$, the MC estimate $\langle k \rangle_{\text{MC}}$, relative error, and total variation distance. The MC mean agrees with the exact mean to within 2% and the TV distance is below 0.04 across all tested L , confirming that the sampler converges to the correct uniform distribution.

Mode convergence study

L	k^*	$k^*/\sqrt{L}$	$k^*/(\sqrt{L} \log L)$	$\langle k \rangle/\sqrt{L}$	$\sigma_k/L^{1/4}$	τ_{int}
20	—	—	—	—	—	—
100	—	—	—	—	—	—
500	—	—	—	—	—	—
2000	—	—	—	—	—	—

(Exact values depend on the run; the key diagnostic is that $k^*/(\sqrt{L} \log L)$ converges toward $0.950 \times \sqrt{6}/(2\pi) \approx 0.370$ from above, and $\langle k \rangle/(\sqrt{L} \log L)$ converges toward $\sqrt{6}/(2\pi) \approx 0.390$, both slowly due to sub-leading corrections of order $\sqrt{L}$.)

Run-length distribution

For each $L \in \{100, 200, 500\}$ the script prints $\langle k \rangle_{\text{MC}}$, $\langle k \rangle_{\text{canon}}$, the canonical α (and the GC fugacity $\pi/\sqrt{6L}$ for comparison), the KL divergences, and a table of $f_{\text{MC}}(j)$, $f_{\text{BE}}(j)$, their ratio, $f_{\text{geom}}(j)$, and its ratio to $f_{\text{MC}}(j)$ for each j . The KL results that feed the paper's §8.3 table are:

L	$D_{\text{KL}}(f_{\text{MC}} \ f_{\text{BE}})$	$D_{\text{KL}}(f_{\text{MC}} \ f_{\text{geom}})$	Ratio
100	≈ 0.0018	≈ 0.080	≈ 45
200	≈ 0.0013	≈ 0.103	≈ 82
500	≈ 0.0010	≈ 0.147	≈ 141

These are single-chain estimates; the paper's jackknife error bars come from `rld_fast.py`, which runs 12 independent chains per L .

(e) What we learn

The uniform measure sampler converges to the correct distribution. Validation against exact enumeration for $L \leq 20$ confirms TV distance below 0.04 and mean error below 2%, establishing that the MH step with the $N_{\text{fwd}}/N_{\text{rev}}$ acceptance ratio correctly targets the uniform measure.

The modal and mean part counts follow $\sqrt{L} \log L$ scaling. The mode convergence study confirms that $k^*/(\sqrt{L} \log L)$ converges toward the Erdős–Lehner coefficient from above, consistent with the exact DP results of `be_saddle.py` and Table 2 of the paper. The naive GC estimate $(\pi/\sqrt{6})\sqrt{L}$ is exceeded by a factor growing as $\log L$, confirming canonical/GC ensemble inequivalence for the part-count observable.

The BE formula fits the run-length distribution substantially better than the geometric. The KL divergence from the BE formula is one to two orders of magnitude smaller than from the geometric, with the ratio growing with L as the bosonic enhancement at small j becomes more pronounced. This is the single-chain version of the result confirmed with jackknife errors in `rld_fast.py`.

The canonical α must be used, not the GC fugacity. The script explicitly solves for α from the exact canonical mean rather than using $\pi/\sqrt{6L}$. Using the GC fugacity as the BE normaliser produces a distribution that sums to $E[k]_{\text{GC}}/E[k]_{\text{canon}} \approx 1/\log L$ times too much probability at each j , making KL divergence computation undefined and the comparison meaningless; this error is avoided by construction.